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In vitro and computational approaches to predict developmental toxicity: Integrating PBPK models with cell-based assays

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A R T I C L E I N F O

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A B S T R A C T

The field of chemical safety assessment is experiencing a transformative shift away from animal testing toward new approach methodologies (NAMs). While *in vitro* screening platforms for teratogenic potential have advanced significantly, their practical application in public health risk assessment has been limited by the lack of quantitative frameworks connecting *in vitro* results to human exposure levels. This study investigated whether prenatal developmental toxicity assays could quantitatively predict developmental toxicity thresholds for nine known reproductive toxicants (bisphenol A, caffeine, carbamazepine, carbaryl, deltamethrin, ethanol, nicotine, thalidomide, and toluene) and one non-teratogen (acetaminophen). We employed a three-tiered physiologically based pharmacokinetic (PBPK) model to translate *in vitro* concentrations from ReproTracker, Stemina DevTOX quickPredict™, and developmental neurotoxicity (DNT) battery assays into human equivalent doses (HEDs). These *in vitro*-derived HEDs were then compared against HEDs based on rodent developmental studies' lowest observed effect levels (LOAELs), as well as human clinical doses and exposure predictions. Our findings demonstrate that the integration of PBPK modeling with Reprotracker, Stemina DevTOX quickPredict™ and DNT assays advances the reduction of animal testing while enhancing human-relevant developmental toxicity assessment. Moving forward requires parallel improvements in our understanding of developmental toxicity mechanisms and computational methods that effectively translate *in vitro* results to human risk evaluation.

1. Introduction

Scientists are developing new *in vitro* testing methods to assess whether chemicals might affect embryonic development and reproduction. Among these approaches, tests using embryonic and induced pluripotent stem cells are particularly promising because they can be scaled up easily and produce consistent results. (Jamalpoor et al., 2022; Palmer et al., 2017; Palmer et al., 2013; Racz et al., 2018; Zhu et al., 2016). One such test is ReproTracker, developed by Toxys (Jamalpoor et al., 2022). This system uses human induced pluripotent stem cells (hiPSCs) as biological indicators to detect potential birth defects from drugs and chemicals. The test works by observing how these stem cells develop into three different cell types: heart muscle cells, liver cells, and early nervous system structures. Scientists evaluate proper development by examining cell appearance and measuring specific biological markers over time. The reliability of ReproTracker has been thoroughly tested using established sets of chemicals (from ICH and EURL ECVAM) that are known to either cause or not cause birth defects, with different mechanisms of action (Augustine-Rauch et al., 2010; Jamalpoor et al., 2022; Seiler et al., 2011; Wilk-Zasadna and Minta, 2009). Some of these results are presented on the ReproTracker website: https://toxys.com/reprotracker/. When identifying compounds that might cause birth defects, ReproTracker achieves 85 % accuracy overall, correctly identifying 85 % of harmful compounds (sensitivity) and 84 % of safe compounds (specificity). The Stemina devTOX quickPredict system is a screening panel based on metabolic changes in embryonic stem cells (Palmer et al., 2017; Palmer et al., 2013; Zhu et al., 2016) and induced pluripotent stem cells (iPSCs) (Palmer et al., 2017) that have been shown to be predictive of teratogenicity and other developmental toxicities. The assay is based on changes in the ratio of ornithine and cystine, two amino acids determined empirically to be perturbed in response to a testing set of 23 pharmaceuticals known to cause a variety of effects in development, including the cardiovascular, craniofacial, central nervous system, limb, skeletal malformation, and embryotoxicity. Other *in vitro* assays deemed ready for regulatory assessment use include the developmental neurotoxicity *in vitro* battery (DNT IVB), which measures changes in key neurodevelopmental processes. The DNT IVB developed for the European Food Safety Authority (EFSA)

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funded research project was based on the OECD/EFSA review of available *in vitro* DNT assays (Fritsche et al., 2017).

While these new testing methods can help predict outcomes relevant for animal studies, they have a key limitation: they can only identify whether a substance might cause birth defects, not determine what doses would be harmful to pregnant women and their developing babies. To bridge the gap, two approaches can be used. Physiologically based pharmacokinetic (PBPK) modeling describes how substances move through the body, helping understand the relationship between exposure and the amount that reaches specific tissues (Clewell et al., 2001; Clewell and Clewell, 2008; Lipscomb et al., 2012). *In vitro* to *in vivo* extrapolation (IVIVE) converts concentrations that show effects in *in vitro* tests to equivalent human exposures (Wetmore, 2015). Altogether, PBPK modeling provides biologically based means of quantitative *in vitro* to *in vivo* extrapolation (IVIVE), allowing translation of outcomes from *in vitro* test methods into a context relevant to human health decisions. For studying effects on development, scientists need specialized PBPK models that account for pregnancy. PBPK models for developmental toxicants require the incorporation of fetal maturation processes and maternal changes during pregnancy, including the placenta and fetal organs (Abduljalil et al., 2012; Kapraun et al., 2019). A three-tiered PBPK model framework was developed in this project. The basic version (Tier 1) focuses on how substances distribute in the pregnant woman's body, without including the fetus. This is suitable for studying very early pregnancy effects, when the drug or chemical concentration in the uterus is most relevant. Even post implantation, maternal plasma levels are often a suitable dose metric. For developmental effects with a window of susceptibility following the formation of the placenta, it will generally be more appropriate to use a model with a fetal compartment that is separated from maternal physiology by the placenta. Therefore, Tier 2 model includes mathematical descriptions of the placenta and fetal compartment. The most significant difference between Tier 1 and Tier 2 PBPK models is the presumed target tissue used for the basis of IVIVE. In Tier 1 models, maternal blood or uterus concentration is expected to be the dose metric. The Tier 2 model has an explicit formulation for the placenta and the whole fetus it supports, which will serve as the target tissue for reverse dosimetry. Finally, Tier 3 models include explicit representations of relevant fetal organ systems. Tier 3 models are appropriate when there is sufficient evidence that developmental endpoints reflect changes during gestation. In practice, the utility of Tier 3 PBPK models is constrained by data availability. Models include mathematical descriptions of fetal tissue compartments, but toxicokinetic data for compound concentrations in these compartments are challenging to collect and are, therefore, incredibly rare. The question of whether maternal plasma levels provide sufficient protection for safety decisions in pregnancy, *versus* implementing higher tier PBPK models, presents a complex scientific and practical dilemma. While more sophisticated PBPK modeling approaches are technically more accurate in representing physiological processes, their practical value must be weighed against the effort required for implementation, especially considering other uncertainties inherent in safety assessments. The progression from screening-level PBPK models (using *in silico* and *in vitro* ADME data) to chemical-specific validated models depends heavily on data availability. However, such data remains exceptionally scarce for pregnant women due to ethical and practical limitations on clinical studies in this vulnerable population. As regulatory frameworks increasingly favor non-animal testing methodologies, the field must develop innovative approaches to integrate *in vitro* and *in silico* data sources effectively.

In this project, we developed a computational PK framework (Fig. 1) that integrates '*in vitro*' inputs from various sources—including the outputs of emerging NAMs—into a flexible model for predicting equivalent whole-body exposures at different life stages. The goal of the 3-tiered PBPK model is to be able to choose the right model designed for the appropriate window of sensitivity for fetal development and evaluate whether prenatal developmental *in vitro* assays PODs can be used as a proxy for *in vivo* developmental toxicity. A recent study by Moreau et al. (2023) showed how combining PBPK models with *in vitro* testing can improve safety assessments without animal testing. Building on this work, the goal of this study was to see if other developmental *in vitro* tests can predict toxic exposure levels using PBPK modeling and IVIVE. The study examined ten different compounds using three different *in vitro* tests (ReproTracker, Stemina DevTOX quickPredictTM, and DNT battery) (See Table 1). IVIVE was used to calculate human equivalent doses (HEDs) that would match the concentrations causing adverse effects in these tests and were then compared to doses known to cause

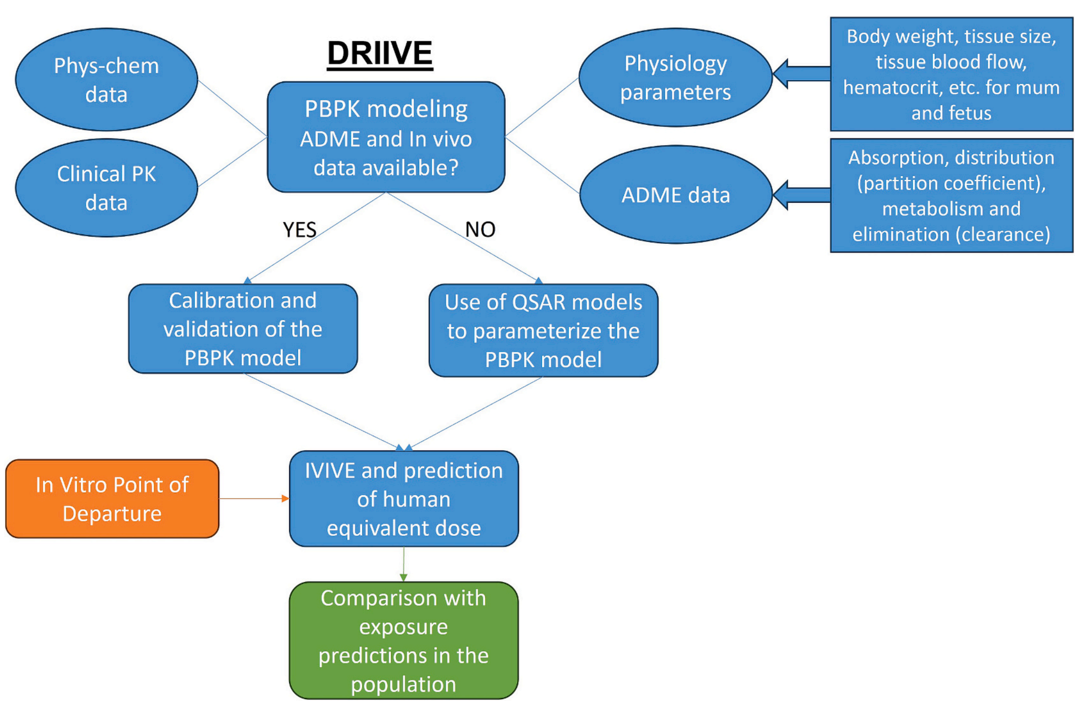

Fig. 1. Developmental and Reproductive *In vitro* to *In Vivo* Extrapolation (DRIIVE) Workflow.

effects in rodent studies, doses used in human medical treatments and predicted human exposure levels.

2. Methods

2.1. Chemical selection

Ten data-rich but structurally different compounds (see Supplemental 1) were selected (acetaminophen CAS: 109–90-2, bisphenol a, caffeine CAS: 58–08-2, carbamazepine CAS: 298–46-4, carbaryl CAS: 63–25-2, deltamethrin CAS: 52918–63-5, ethanol CAS: 64–17-5, nicotine CAS: 54–11-5, thalidomide CAS: 50–35-1 and toluene CAS: 108–88-3). Each compound selected for the case study meets the following criteria: 1) tested in dose-response in ReproTracker, devTox quickPredict, or any other assay, 2) has credible available metabolism data, and 3) is associated with developmental effects in in-life studies. Chemicals were chosen to include mechanisms that induce developmental toxicity in different gestation ages, as well as some negative controls. Chemical selection considered distinct windows of sensitivity for fetal development, ensuring that compounds were selected to represent both more specific mechanisms of toxicity encountered throughout organogenesis and generic toxicity in early fetal development.

2.2. PBPK model description

The tier 1 PBPK model has compartments for blood and six tissue compartments: gastrointestinal (GI) tract, liver, fat, uterus, skin, and rest of the body. The tier 2 and 3 PBPK models have maternal compartments for blood and six tissue compartments: gastrointestinal (GI) tract, liver, fat, skin, placenta, and rest of the body. The tier 2 model also integrates the fetal body with the amniotic fluid whereas the tier 3 model integrates the fetal body as multiple compartments (Liver, brain, and the rest of the body) with the amniotic fluid. Each model incorporates age-dependent human physiology of pregnant women.

Table 1
In vitro point of departures.

Chemicals	Reprotracker	Stemina DevTOX quickPredict™	DNT *in vitro* battery[c]
Acetaminophen	NT	ND (max conc = 300 μM)	Negative
Caffeine	NT	696.37 μM	–
Nicotine	NT	727.36 μM	11.8 μM (Synaptogenesis)
Bisphenol A	NT	ND (max conc = 100 μM)	74.4 (neurite initiation hN2) 11.9 μM (UKN2)
Carbamazepine	7.8 μM	1.09 μM	–
Carbaryl	NT	2.92 μM	5 μM (UKN5) 15 μM (UKN4)
Thalidomide	0.4 μM	0.09 μM	40.9 μM (hN2) 0.6 μM (NPC5)
Deltamethrin	NT	NT	3.9 μM (h-neuronal network formation) 112.8 μM (UKN5)[d]
Ethanol[a]	NT	NT	NT in DNT *in vitro* battery 250 mM (neurite outgrowth)
Toluene[b]	NT	NT	NT in DNT *in vitro* battery 250–1000 μM (Astrocytes)

NT: not tested; ND: not defined.
[a] Ethanol – Zink et al. (2020)
[b] Toluene– Burry et al. (2003)
[c] Masjosthusmann et al. (2020), Aschner et al. (2017); Harrill et al. (2018)
[d] non-specific hits (the benchmark response (BMC) for the effect is not separated from the BMC for cytotoxicity/cell viability).

The current models (Tier 1, 2 and 3) can simulate how substances move through the body when taken by different routes - oral, inhaled, dermal, or intravenous - for both single and repeated daily exposures. They account for how substances are metabolized in the mother's liver and blood, though they don't include metabolism in the fetus. These models use a “flow-limited” approach for most tissues, meaning they assume: 1- Blood flowing through a tissue is the main factor controlling how substances move into that tissue and 2- Substances quickly reach an equilibrium between blood and tissue concentrations. The only exception is deltamethrin, where the model treats movement into fat tissue as being limited by diffusion rather than blood flow.

A schematic of the model structure is shown in Fig. 2. Berkeley Madonna software (Version 10.5.1) was used to run simulations, with the complete model code provided in supplemental 2.

2.2.1. Parameter estimation and analysis

The models draw on several key data sources for their parameters. Most pregnancy-related parameters came from Kapraun et al. (2019), and uterus-specific values came from Abduljalil et al. (2012). Both sources provide detailed data about how various body measurements (tissue volumes, blood flow rates…) change during pregnancy. Male physiological data were also used, as some of the original studies used to verify PBPK model accuracy were conducted in men. In these cases, PBPK models were first developed and validated using male data. They were then adapted for pregnancy by accounting for physiological changes while keeping certain drug-specific parameters the same to predict limited pharmacokinetic data in pregnant women. The male physiological data came from published life-stage models by Ruark et al. (2017), Song et al. (2016); Wu et al. (2015), covering factors like body weight, blood flow, and tissue measurements. All female and male physiological parameters used in the three tier models are summarized in Moreau et al. (2025).

Chemical specific parameters, including clearance, oral absorption rates and partition coefficients, are all described in Moreau et al. (2025).

2.2.2. Calibration and validation of the model

The performance of the models was evaluated using *in vivo* pharmacokinetic studies for the ten compounds in humans and results are presented in Moreau et al. (2025). Following World Health Organization (WHO, 2010) criteria, predictions of maximal concentration (Cmax) that were within a factor of two of the experimental data were deemed adequate.

2.2.3. Global sensitivity analysis

The global sensitivity was conducted in the Rvis software (version RVis_v0.9.11068.1_x64 from https://github.com/GMPtk/RVis was used) according to McNally et al. (2011). The description of the method used as well as the results are presented in Moreau et al. (2025).

2.3. In vitro to in vivo extrapolation (IVIVE)

2.3.1. In vitro point of departures

The aim of this study was to assess whether an *in vitro* prenatal developmental toxicity assay could predict relevant *in vivo* developmental toxicity exposure dosimetry of drugs and chemicals using PBPK models, along with an IVIVE approach. Specifically, we applied IVIVE to predict the human equivalent developmental toxicity dose levels (HEDs) for nine known human reproductive toxicants—caffeine, thalidomide, carbamazepine, deltamethrin, carbaryl, toluene, ethanol, bisphenol A, and nicotine—and one negative control, acetaminophen. These predicted HEDs correspond to *in vitro* concentrations linked to adverse outcomes in the ReproTracker, Stemina DevTOX quickPredict™, and DNT *in vitro* testing battery (OECD, 2023). We then compared the *in vitro*-derived HEDs to those based on the Points of Departure (PODs) from developmental studies in rodents, as well as human clinical doses or exposure predictions for the human population.

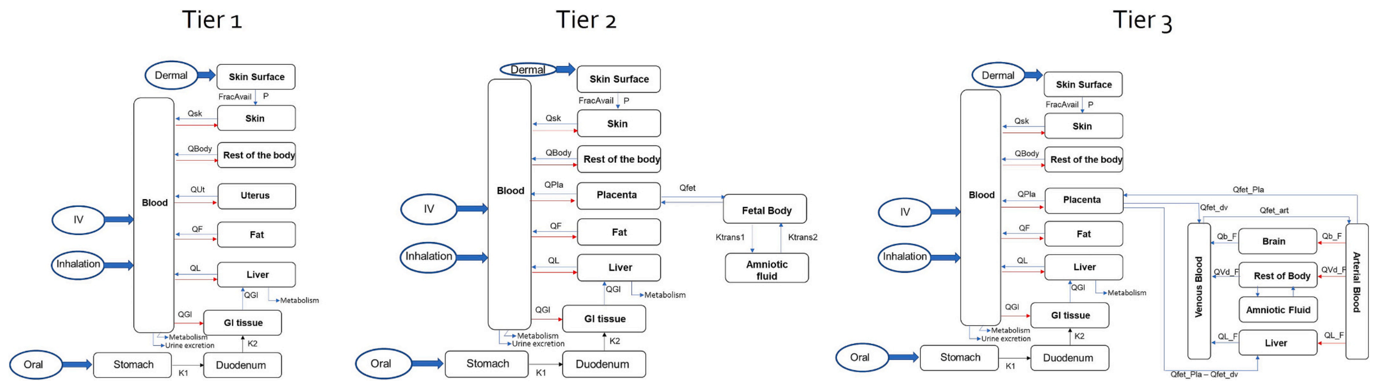

Fig. 2. Structure of the Three-Tier PBPK model. QL, QGI, QBody, QF, Qsk, and Qpla refer to blood flow to each maternal tissue compartment. Fa describes the fraction available for absorption. K1 and k2 represent absorption rates for lumen to duodenum and from duodenum to GI tract. Ktrans1 and 2 represent the transfer rates in and out of the amniotic fluid. Qfet_Pla, Qfet_dv, Qfet_art, Qb_F, QVd_f, and QL_F refer to blood flow to each fetal compartment. Qfet_Pla is the fetal blood flow rate through the placenta. The ductus venosus effectively channels a portion of the blood flowing through the umbilical vein directly to the venous system, while the remainder is directed to the liver. Consequently, the blood flow through the ductus venosus (Qfet_dv) is always lower than the blood flow to and from the placenta (Kapraun et al., 2019).

The Stemina DevTOX quickPredict™ and Toxys ReproTracker are screening assays used to detect the teratogenic potential of compounds. Neurodevelopment *in vivo* is a complicated process involving various cellular mechanisms. Consequently, a wide range of *in vitro* tests that simulate and measure impacts on various neurodevelopmental processes may be able to identify substances that may influence brain development at various stages of development (Harrill et al., 2018). According to Aschner et al. (2017), this suggested DNT battery specifically

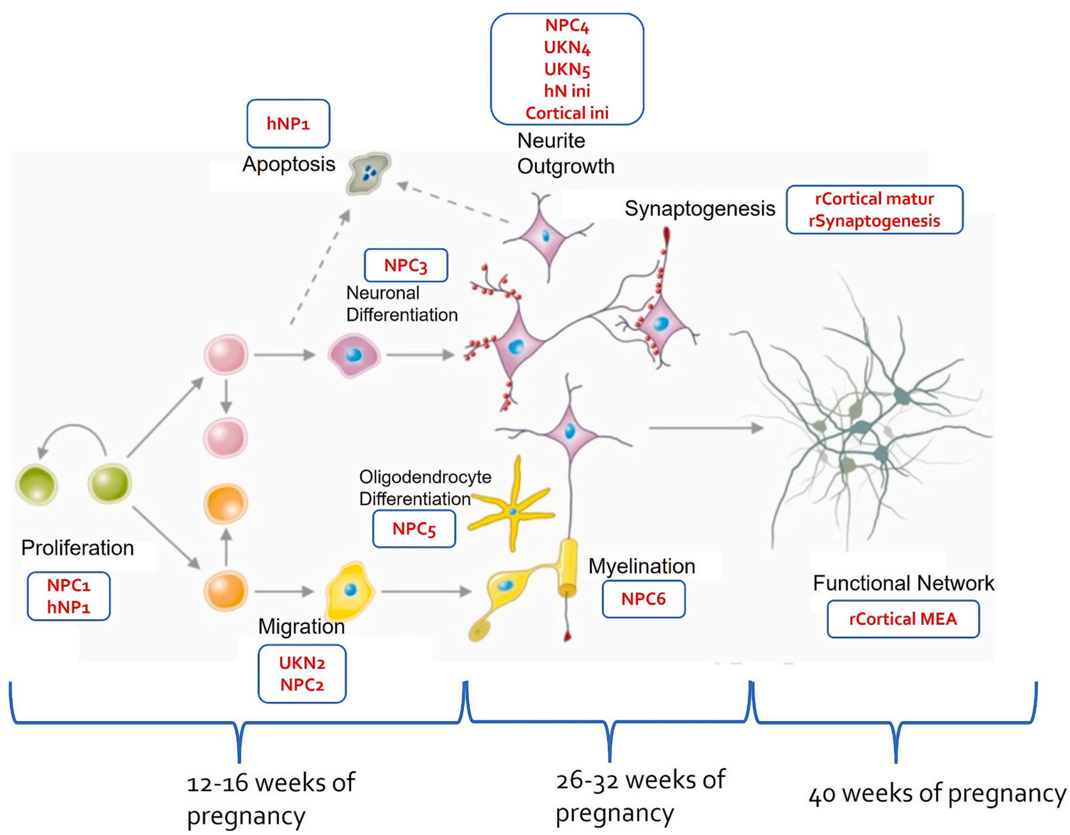

Fig. 3. Key Neurodevelopmental Processes modified from Aschner et al. (2017). Neural stem cells (NSCs, green) proliferate and differentiate into various neural progenitor cells (NPCs), including neuronal progenitors (light purple) and glial progenitors (orange). These progenitor cells further proliferate, migrate, and mature into neurons (purple) and glial cells (yellow). As maturation progresses, neurons extend neurites and establish synapses (red), while excess cells undergo apoptosis (grey). When these processes occur in a synchronized manner, cell-cell interactions form a functional neuronal network (olive). *Abbreviations*: h-human; r-rat; matur-maturation; ini-initiation; MEA-microelectrode array. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)
NPC1: Proliferation assay of primary human neuronal progenitor cells (hNPCs).
NPC2: Migration assay of primary hNPCs.
NPC3–4: Neuronal differentiation assay (NPC3) and evaluation of neuronal morphology (neurite length and area) in young neurons derived from hNPCs.
NPC5: Oligodendrocyte differentiation assay.

addresses several important neurodevelopmental processes, including proliferation, differentiation, migration, neurite outgrowth, synaptogenesis, and the construction and function of neural networks. These endpoints were selected based on well-established neurodevelopmental processes. The assays included in the DNT *in vitro* battery are highlighted in red in Fig. 3. For more details on these assays, please refer to Masjosthusmann et al. (2020). The establishment of functioning neuronal networks represents the culmination of neurodevelopment. Neural network coordination and connectivity are essential for basic cognitive functions, including learning, memory, and attention. It may therefore be useful to find potential developmental neurotoxicants using an experiment that measures the successful creation of brain networks. To achieve this, Frank et al. (2017) created a network formation test that uses primary rat cortical neurons' microelectrode array (MEA).

2.3.2. *In vivo point of departures*

Allometric scaling based on body weight was applied to convert the Points of Departure (PODs) from rodents into human equivalent doses (HEDs), allowing for comparison with the HED derived from the *in vitro* reproductive assay (Eq. 1). To calculate the HED, the dose adjustment factor (DAF) is multiplied by the animal exposure (in mg/kg-day), yielding the human equivalent exposure (in mg/kg-day). The calculation of the DAF is outlined in Eq. 2 (EPA, 2013).

$$\text{HED (mg/kg)} = \text{Animal POD (mg/kg)} \times \text{DAF} \quad (1)$$

$$\text{DAF} = \left(\text{Weight}_{\text{animal}}\ (\text{kg})/\text{Weight}_{\text{human}}\ (\text{kg})\right)^{1/4} \quad (2)$$

2.3.3. *In vitro to in vivo extrapolation*

In vitro developmental PODs were used to derive Human Equivalent Doses (HEDs) using the generic PBPK models developed in this project. The choice of the tier PBPK model used to do IVIVE depends on the *in vitro* prenatal developmental toxicity assay used. Biomarkers utilized in the ReproTracker or the Stemina DevTOX quickPredict™ occur very early on in the pregnancy process, which is why the tier 1 model—containing only the uterus and excluding the placenta and fetus—was used. For the DNT *in vitro* testing battery, simulations were run in the tier 1, 2, or 3 PBPK model, depending on the assay used and according to OECD (2023) (Fig. 3). The tier 1 model was run at 12 weeks of pregnancy, and tier 2 and 3 models were run at 16, 26, 32 or 40 weeks of pregnancy to compare results between outputs of *in vitro* models.

Monte Carlo simulations were employed to estimate the median, as well as the 5th and 95th percentiles, of concentrations in the uterus, fetus, and maternal plasma for individuals exposed to a fixed dose of 1 mg/day. Individuals in the upper 95th percentile exhibit higher plasma or uterus/fetus concentrations under the same exposure conditions, representing a sensitive population (Fig. 4). Exposure scenarios for each chemical are detailed in Table 2. Additionally, reverse dosimetry was used to predict administered doses corresponding to *in vitro* active concentrations determined from reproductive assays.

3. Results

3.1. *PBPK modeling*

All the simulation results can be found in Moreau et al. (2025). Globally, the models predicted median plasma concentration time courses that aligned well with observations across all datasets, with predicted median concentrations differing from observed values by less than a factor of two (Fig. 5). The results also demonstrated that the model consistently captured the overall trends in the data. For a few chemicals, the fitting was more complex. For BPA, the model predictions underestimated the BPA plasma concentrations (Moreau et al., 2025). However, this can be explained by the result of contamination in the sample collection-processing chain, not actual exposure. In the case of deltamethrin, no human *in vivo* data was available, so the human PBPK model was not validated in human. However, a rat PBPK model was developed, calibrated and validated with *in vivo* rat PK data (Data not shown). Rats partition coefficients and scaled clearances were used in

Table 2
Scenario of exposure.

Chemical	Route of exposure	Scenario of exposure
Carbamazepine	Oral	Dose twice daily until a steady state
Thalidomide	Oral	Bolus dose
Acetaminophen	Oral	Bolus dose 3 times a day[a] until steady state
Caffeine	Oral	Bolus dose 3 times a day[a] until steady state
Nicotine	Inhalation	2.3 s duration puff, 30s puff interval, 10 puffs per session, 15 sessions
Bisphenol A	Oral	Bolus dose 3 times a day[a]
Ethanol	Oral	Infusion (1 glass of alcohol over 15 min every day)
Toluene	Inhalation	7 h per day
Carbaryl	Oral	Bolus dose 3 times a day[a]
Deltamethrin	Oral	Bolus dose 3 times a day[a]

[a] Medication taken at breakfast, lunch, dinner.

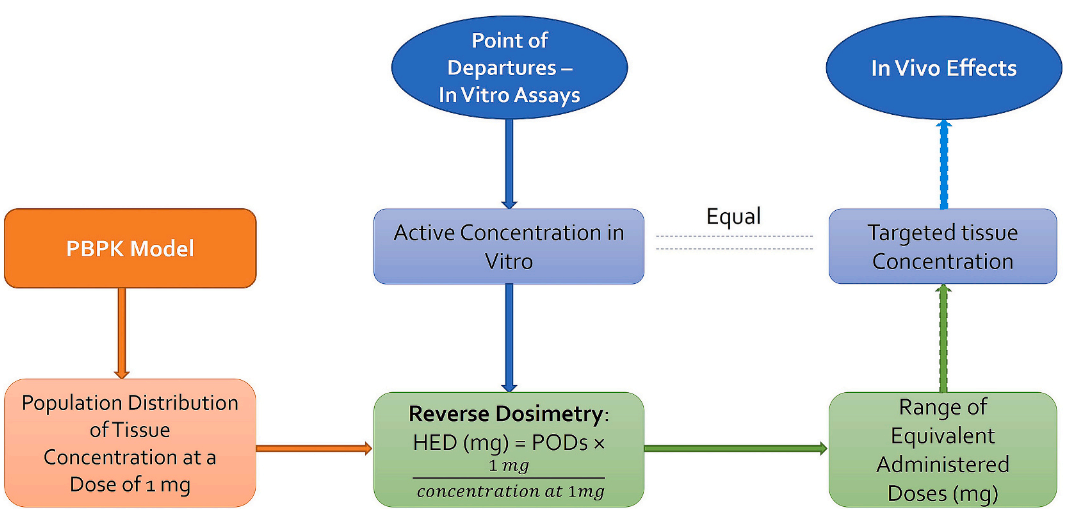

Fig. 4. *In vitro* to *in vivo* extrapolation: Physiological parameters, along with ADME-related data describing chemical behavior within the system, were utilized to construct a PBPK model. This model predicts population distributions of plasma concentration for any given daily dose. Reverse dosimetry allows estimation of administered doses equivalent to *in vitro* active concentrations, enabling comparisons with *in vivo* measurements (Adapted from Bell et al. (2018)).

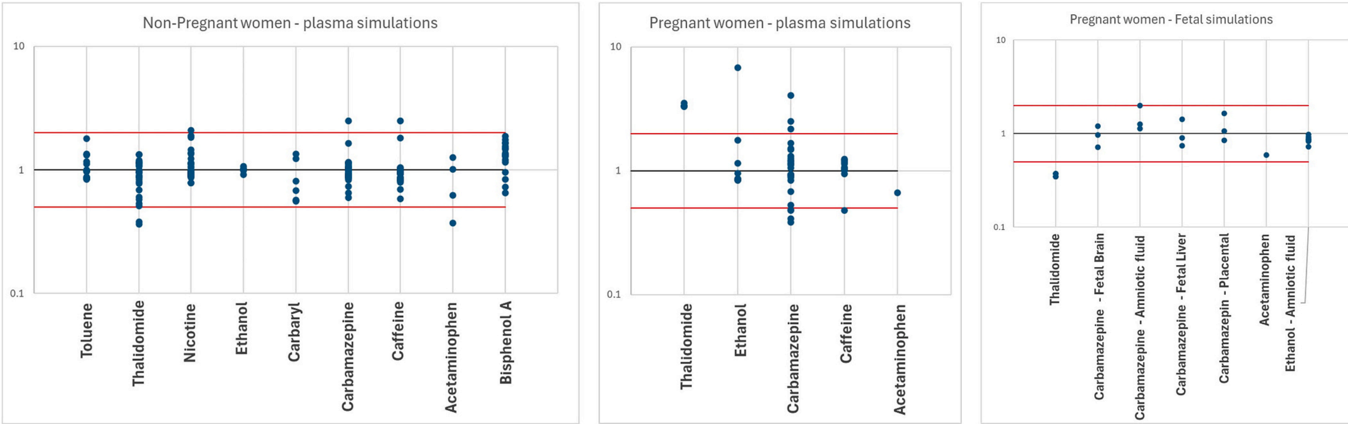

Fig. 5. Fold difference between measured and predicted concentrations in non-pregnant and pregnant women.

Table 3
Animal PODs and their HEDs.

Compounds	Species[a,b]	POD/ effect	POD (mg/kg)	DAF	HED (mg/kg)	UF	Final HED (mg/kg)	Final HED (mg)	Human exposure
Acetaminophen		Negative Control							3000 mg
Caffeine	Rat	NOAEL[b] Teratogenicity	>205	0.247	50.64	30	1.69	>113.3	300 mg
Nicotine	Rat	NOAEL[c] Developmental toxicity	15	0.247	3.71	30	0.12	8.29	15 mg
Bisphenol A[l]	Mice	LOAEL[d] Reproductive, developmental toxicity (reduced litter size)	1250	0.139	173.8	30	5.79	388.6	3.06E-5 - 3.31E-5 mg/kg
	Rat	NOAEL[e] Developmental neurotoxicity	10	0.247	2.47	30	0.08	5.52	
Carbaryl[l]	Rat	LOAEL[e] Decreased fetal body weight and incomplete ossification of multiple bones	30	0.247	7.41	30	0.25	16.57	4.13E-7 - 5.12E-5 mg/kg
	Rabbit	NOAEL[e] Evidence of teratogenicity at maternally toxic dose	150	0.478	71.7	30	2.39	160	
	Rat	NOAEL[f] Reproductive and developmental toxicity	5	0.247	1.235	30	0.04	2.76	
Deltamethrin[l]	Rat	NOAEL[g] Developmental neurotoxicity	16.1	0.247	3.98	30	0.13	8.89	1.28E-7 - 7.88E-7 mg/kg
	Rabbit	NOAEL[f] Reproductive and developmental toxicity	32	0.478	15.3	30	0.51	34.21	
Ethanol[k]	Rat	NOAEL[g] Developmental toxicity, significant effect on ossification	5200	0.247	1284	30	42.81	2873	4.07E-5 - 5.93E-3 mg/kg
	Rat	LOAEC[g] Decreased pup weight and delayed skeletal ossification	1000	0.247	247	30	8.23	–	
Toluene[i,j]	Rat	LOAEC[h] Developmental neurological effects	1200	0.247	296.4	30	9.88	–	5.77E-5 - 2.26E-2 mg/kg
	Mice	LOAEC[h] Skeletal retardation and decreased pup weight	265	0.139	36.84	30	1.23	–	
Thalidomide	Rat	LOAEL[j] Abortions	50	0.247	12.35	30	0.41	27.51	50 mg
	Rabbit	LOAEL[k] Decreased number of fetuses and increased resorption	60	0.478	28.68	30	0.96	64.15	
Carbamazepine	Rat	LOAEL[j] Decreased implantation sites	400	0.247	98.8	30	3.29	221	200 mg
	Rabbit	LOAEL[j] Resorptions	225	0.478	107.6	30	3.59	241	

[a] Rat, rabbit, mice, and human body weight were 0.25, 3.5, 0.025, and 67.1 kg, respectively.
[b] Collins et al. (1983); Collins et al. (1987) and NTP (1984).
[c] ECHA (n.d.)
[d] FDA, 2007.
[e] Canada (2009).
[f] EFSA (2009)
[g] ECHA.
[h] EPA (2005), Hass et al. (1999) and Ungvary and Tatrai (1985)
[i] ppm.
[j] Christian et al. (2007), Schardein and Keller (1989), Newman et al. (1993).
[k] Human population exposure predictions (ExpoCast) – Median – 95th percentile.
[l] NHANES median – upper bound human exposure prediction for women of reproductible age.

humans and simulations were run at 0.01 mg/kg in non-pregnant female and compared with simulation with the model developed in Mallick et al. (2020a); Mallick et al. (2020b). Maximal female plasma concentration simulation with the tier 1 model was 8.13 μg/L at steady state (336 h). With the PBPK model developed by Mallick et al. (2020a); Mallick et al. (2020b), the female maximal plasma concentration at steady state (336 h) was 10.75 μg/L, which is close to the simulations from the tier 1 model. For more details on all the simulations, see Moreau et al. (2025).

Globally, the sensitive parameters seem to be similar between the 10 compounds analyzed (Moreau et al., 2025). When we look at the physiological parameters, Vffm (fat free mass) is sensitive in all compounds as well as the liver volume (except in the case of thalidomide). Parameters related to uterus, or fetal development are the uterus volume and blood flow, which are low sensitive for DLM, carbamazepine, caffeine, and nicotine, the fetal brain volume as well as the fetal arterial flow, which is only low sensitive for toluene, and the fetal hematocrit, which is low sensitive in most compounds except for thalidomide, carbamazepine and bisphenol A. As expected, alveolar ventilation is low to medium sensitive for nicotine and toluene. The fat volume is also low sensitive for four compounds (carbamazepine, nicotine, acetaminophen and toluene).

Intrinsic clearance is sensitive for all compounds except for toluene and nicotine. The sensitivity spans from low-medium (carbaryl), to medium (carbamazepine, DLM and bisphenol A), to medium high (caffeine, acetaminophen, ethanol, and thalidomide). After inhalation exposure, toluene and nicotine are quickly and highly metabolized explaining why liver metabolism is not a sensitive parameter.

The main partition coefficients that are sensitive are the rest of the body, the uterus, the placenta and the fetal brain PC. They are mostly medium sensitive to all compounds.

3.2. *In vitro to in vivo extrapolation (IVIVE)*

When converting animal study doses to human equivalent doses (HEDs), several factors need to be considered. First, we convert the animal dose to a human equivalent using the Dose Adjustment Factor (DAF) (EPA, 2013) (Table 3). This factor accounts for differences in body weight between species (see eqs. 1 and 2). However, we need to account for uncertainties using two uncertainty factors. The first one is the uncertainty factor for interspecies differences (UFA) which accounts for toxicokinetic and toxicodynamic differences between species (rodents *versus* humans). We only use 3 instead of the full 10 because the DAF already handles toxicokinetic differences. The second uncertainty factor is for intraspecies differences (UFH) which accounts for variations in sensitivity among different human subpopulations. The total Uncertainty Factor (UF) is therefore:

$$\text{Total UF} = \text{UFA} \times \text{UFH} = 3 \times 10 = 30 \quad (3)$$

The final HED in mg/kg is calculated as:

$$\text{Final HED (mg/kg)} = \text{POD} \times \text{DAF}/30 \quad (4)$$

To get the absolute dose in mg, this final HED can be multiplied by the human's body weight.

A generic Physiologically Based Pharmacokinetic (PBPK) model was developed to determine Human Equivalent Doses (HEDs) from *in vitro* point of departures for ten substances. The process involved several key steps. First, the model used Monte Carlo simulations to analyze how these substances would distribute in the human body. Using a standardized 1 mg/day dose, the simulations predicted the concentrations in both plasma and uterine/fetal tissue across a population. The model then employed reverse dosimetry to determine what administered doses would produce concentrations matching those found to be active in three different *in vitro* developmental toxicity assays: ReproTracker, Stemina DevTOX quickPredict™, and DNT. This approach allowed for the conversion of *in vitro* No Observed Adverse Effect Levels (NOAELs) and Lowest Observed Adverse Effect Levels (LOAELs) into human doses.

The inclusion of the 95th percentile data was particularly important, as it represented individuals who would experience higher concentrations of these substances in their plasma or uterine tissue despite receiving the same dose as others. These individuals effectively represent a sensitive subpopulation that requires special consideration in safety assessments. Table 4 shows the results of IVIVE.

4. Discussion

Emerging screening technologies offer promising approaches for predicting developmental toxicity without relying on animal studies. Specialized human-based assays like ReproTracker, Stemina DevTOX quickPredict™, and the DNT battery have demonstrated effectiveness in evaluating teratogenic and developmental neurotoxicity potential of pharmaceuticals and chemicals, showing strong correlation with traditional testing systems. However, integrating these screening platforms into safety frameworks remains challenging, particularly when quantifying compound-specific dose-response relationships at critical developmental stages. The key challenge lies in determining precise exposure levels at target sites during specific developmental windows of susceptibility.

The goal of this study was to evaluate whether these developmental *in vitro* assay PODs can be used as a proxy for *in vivo* developmental toxicity. This was accomplished by using human PBPK models to simulate internal exposures equivalent to the *in vitro* POD exposure values and comparing them to known experimental studies. In other words, we implemented a PBPK model to analyze how nine known teratogenic compounds move through and interact with biological systems. This model allowed us to perform IVIVE, effectively translating *in vitro* results into predictions of human exposure.

DART (Developmental and Reproductive Toxicity) studies required by regulatory agencies like FDA and EMA aim to assess how pharmaceuticals affect mammalian reproduction to inform human risk assessment. Current guidelines mandate testing in both rodent and non-rodent species (typically rabbits) to account for potential species-specific responses to drugs. While traditional animal-based toxicology screening has been the historical standard, computational modeling and alternative screening methods are increasingly accepted within toxicology and pharmaceutical development. Standard risk assessment relies on determining Points of Departure (POD) from animal studies and deriving human reference doses through uncertainty factors that compensate for data gaps and variability. Our case study employed allometric scaling to derive human equivalent doses from rodent studies, though a rodent PBPK model would have provided more refined predictions. While PBPK modeling offers superior accuracy, we chose a simpler approach to establish proof of concept. To address species differences (extrapolation from rat and rabbit to humans), we applied an interspecies extrapolation factor (UFA). The DAF was implemented to address toxicokinetic variations between species, though it's important to note that standard DAFs typically don't account for toxicodynamic response components. Species-specific toxicodynamic differences affect toxicity magnitude, justifying retention of the toxicodynamic component (approximately 3) of the UFA, as per EPA guidelines (EPA, 2013). The analysis also incorporated an intraspecies uncertainty factor (UFH) of 10 to protect sensitive human populations, resulting in a total uncertainty factor of 30 applied to rat and rabbit LOAELs and NOAELs. A significant challenge in developmental studies and IVIVE arises when considering developmental windows of susceptibility and exposure duration. The inherent limitations of alternative testing systems, which cannot replicate extended exposure periods of weeks or months, create a fundamental constraint in the IVIVE approach, particularly for developmental toxicity assessment.

In our model, we used a simplified approach to represent how chemicals move across the placenta, relying primarily on passive partitioning mechanisms. While this basic representation proved adequate

Table 4
In vitro to *in vivo* extrapolation (IVIVE) and comparison with *in vivo* HED – median (5th percentile – 95th percentile).

Chemicals HEDs	*In vitro* assays	*In vitro* PODs Table 2 (μM)	Maternal Plasma[1]	Uterus[a] for tier 1 Fetus for tier 2 Fetal blood (Bl) and brain (Br) for tier 3
Acetaminophen (mg/day)	Stemina DevTOX quickPredict™	300	10,014 (6776–14,727)	13,011 (7031–23,234)
Caffeine (mg/day)	Stemina DevTOX quickPredict™	696.37	8782 (6375–12,110)	15,569 (9347–26,191)
Nicotine (mg/day)	Stemina DevTOX quickPredict™	727.36	40,559 (30,307–54,581)	20,141 (11893–37,294)
	DNT-tier 2 (32 W)	11.8	603.2 (406.3–805.1)	499.5 (303–852.9)
	DNT-tier 3 (32 W)	11.8	591 (378–784)	Bl: 515 (330.4–827) Br: 579.3 (317.3–1078)
Bisphenol A (mg/kg/day)	Stemina DevTOX quickPredict™	100	6302 (2978–16,505)	8750 (3810–23,937)
	DNT-tier 1 (12 W)	11.9	720.73 (390.8–1333)	402 (219–745)
	DNT-tier 2 (26 W)	74.4	3587 (1997–7005)	2228 (1079–4969)
	DNT-tier 3 (26 W)	74.4	3763 (2089–7192)	Bl: 1983 (986–4257) Br: 813 (352–1936)
Carbaryl (mg/kg/day)	Stemina DevTOX quickPredict™	2.92	13.24 (7.48–22.70)	43.44 (21.41–87.69)
	DNT-tier 2 (26 W)	5	18.80 (10.16–33.32)	66.81 (31.49–133.15)
	DNT-tier 3 (26 W)	5	18.30 (10.52–33.57)	Bl: 68.99 (36.06–138.7) Br: 71.45 (32.91–159.7)
	DNT-tier 2 (26 W)	15	56.41 (30.47–99.95)	200.4 (94.46–399.5)
	DNT-tier 3 (26 W)	15	54.91 (31.56–100.7)	Bl: 207 (108.2–416.2) Br: 214.4 (98.72–479.1)
Deltamethrin (mg/kg/day)	DNT-tier 2 (16 W)	0.6	0.73 (0.49–1.11)	0.27 (0.15–0.48)
	DNT-tier 2 (26 W)	112.8	138.3 (87.45–203.7)	52.63 (28.66–88.19)
	DNT-tier 3 (26 W)	112.8	135.1 (89.56–199.1)	Bl: 53.92 (28.88–97.66) Br: 328.7 (155.5–679.5)
	DNT-tier 2 (40 W)	3.9	4.51 (2.91–6.79)	1.70 (0.94–2.97)
	DNT-tier 3 (40 W)	3.9	4.56 (2.84–6.62)	Bl: 0.98 (0.54–1.82) Br: 9.86 (4.85–20.65)
Ethanol (mg/kg/day)	Literature-tier 2 (26 W)	250,000	10,351 (6263–16,605)	16,566 (9022–31,333)
	Literature-tier 3 (26 W)	250,000	10,576 (6570–16,179)	Bl: 14,923 (8339–26,983) Br: 28,939 (14,080–60,408)
Toluene (ppm)	Literature-tier 2 (16 W)	250	1989 (1470–2614)	634.3 (383.7–1049)
	Literature-tier 3 (16 W)	250	1970 (1481–2635)	Bl: 469.8 (342.6–630.2) Br: 66.25 (39.53–110.9)
Thalidomide (mg/day)	Stemina DevTOX quickPredict™	0.09	2.38 (1.59–3.51)	6.91 (3.74–12.70)
	ReproTracker	0.4	10.06 (6.45–16.34)	30 (19.04–49.44)
	DNT-tier2 (26 W)	40.9	1110 (751–1658)	3682 (2046–6909)
	DNT-tier 3 (26 W)	40.9	1112 (757.6–1715)	Bl: 2851 (1606–5066) Br: 10,723 (5300–23,207)
Carbamazepine (mg/day)	Stemina DevTOX quickPredict™	1.09	17.29 (16.01–18.48)	22.47 (14.44–34.18)
	ReproTracker	7.8	98.11 (64.94–149.4)	139.1 (91.88–210.5)

[a] Human equivalent doses are derived using the PBPK model and assuming the LOAEL concentration from the *in vitro* assays are plasma or uterine concentrations.

for our purposes, it's important to acknowledge its limitations. For some chemicals, active transport processes involving specific placental transporters may play a significant role in their transfer across the placental barrier. However, given that the specific roles and mechanisms of these transporters remain poorly understood for many chemicals, we opted to maintain a straightforward modeling approach rather than incorporate speculative transport mechanisms. This simplified modeling decision reflects the current state of scientific knowledge while acknowledging that future research may reveal more complex placental transport processes for certain compounds.

The complexity of IVIVE PK models might vary. Basic one-compartment models, which assume linear dose-response and steady-state kinetics, can be rapidly parameterized using clearance and plasma protein binding data from *in vitro* studies (Wetmore, 2015; Wetmore et al., 2012). For developmental toxicity, pregnancy-specific PBPK models that account for *in-utero* exposure are preferred, though these require modeling complex maternal-fetal tissue interactions. Our initial (Tier 1) PBPK model specifically addresses compound distribution in pregnant women during early pregnancy, focusing on the period before fetal development. This approach directly corresponds with the endpoints measured by ReproTracker and Stemina DevTOX quickPredict™ assays, which examine cellular events during critical early developmental stages of gastrulation and organogenesis. The model concentrates on the first five weeks of pregnancy, a period that precedes placental formation but encompasses the beginning of organogenesis (which extends from weeks three through eight). During this crucial developmental window, plasma and uterine chemical concentrations serve as the primary dose metrics for assessment. This timing aligns with established developmental biology, as documented by Burton and Jauniaux (2018). Pregnancy presents unique modeling challenges due to significant physiological changes that occur throughout gestation. While traditional PBPK models often employ static parameters, the substantial anatomical changes in both mother and developing fetus necessitate a more dynamic modeling approach. For analyzing developmental endpoints beyond this early window, more sophisticated modeling frameworks (designated as Tier 2 and 3) would be required to accurately represent maternal-fetal transport mechanisms and interactions. This tiered approach allows us to match the complexity of our modeling strategy with the specific developmental stage under investigation, ensuring appropriate analysis while maintaining model efficiency.

As regulatory frameworks shift toward animal-free testing approaches, the generic PBPK model developed in this case study offers broader applicability. Using Quantitative Structure-Activity Relationship (QSAR) models to predict ADME (absorption, distribution, metabolism, and excretion) properties, this modeling framework could be extended to evaluate other chemicals when experimental data is unavailable. Pregnancy is associated with several physiological changes, including metabolic clearance, which can alter the pharmacokinetics (PK) and pharmacodynamics (PD) of compounds. PBPK pregnancy models adequately account for many physiological changes in pregnancy (*e.g.*, increased plasma volume, decreases in serum albumin, increases in glomerular filtration rate (GFR), changes in blood flows, organ and tissue volume changes and growth of a placenta and fetus). However less is known about metabolic clearance of drugs and chemicals or the potential involvement of transporter proteins, both of which may be under the influence of hormones produced during pregnancy. Our modeling approach utilized adult male or fitted metabolic parameters for drug-specific modeling, despite documented evidence that pregnancy significantly affects drug metabolism. Research by Darakjian and Kaddoumi (2019) and Isoherranen and Thummel (2013) has shown that pregnancy induces differential changes in drug-metabolizing enzymes. For instance, while CYP3A4 enzyme activity increases during pregnancy, CYP1A2 activity decreases. Although these metabolic alterations could potentially impact pharmacokinetics, our model did not explicitly account for these pregnancy-specific changes. Instead, we employed established adult male parameters values to simulate and predict pharmacokinetic patterns in pregnant women. In some simulations, clearance data were fitted to the *in vivo* data, taking into consideration the change of clearance during pregnancy. This simplification represents a limitation in our current modeling framework that should be considered when interpreting the results. The placenta and the fetus liver contain metabolic enzymes and transporter proteins. The maternal hepatic CYP and UGT isoenzymes levels are much greater abundance than in the placenta and fetal hepatic tissue and in many cases placenta and fetal metabolism are expected to have a minimal impact on whole body maternal pharmacokinetics (Robinson et al., 2020). This was not taken into consideration in this work although accounting for fetal metabolism, placenta metabolism and transport mechanisms (diffusion and protein transporters) may be important in a case-by-case situations for a drug or chemical.

In vitro pharmacokinetic consideration is also important. Compounds can bind to various elements including plastics, serum proteins, and agarose support, or be affected by evaporation. While the free concentration of a chemical is crucial for determining both kinetics and dynamics, our analysis did not account for these factors in the quantitative *in vitro* to *in vivo* extrapolation (QIVIVE). More precise results could potentially be achieved by incorporating mass-balance models, such as those developed by Armitage et al. (2021) and Armitage et al. (2014), to calculate free medium concentration.

Our PBPK model demonstrated strong predictive capability across four test compounds (Fig. 6). For comparison between the HED from *in vitro* assays and *in vivo* experiments, we decided to use the *in vivo* HED without the uncertainty factors. For thalidomide, the model-derived Human Equivalent Doses (HEDs) ranged from 6 to 49 mg/day, closely matching clinical teratogenic doses (50 mg/day) and being under HEDs derived from animal studies (829–1924 mg/day). Carbamazepine showed more conservative results, with model-predicted HEDs (17.29–139.1 mg/day) falling below animal-derived HEDs (6629–7220 mg/day). In the DNT battery assay for thalidomide, the observed ratio with the *in vivo* assay ranges from 1 to 13. This ratio doesn't indicate a lack of sensitivity in the *in vitro* assay. Instead, it reflects that the *in vivo* assay primarily demonstrated teratogenicity rather than specific neurodevelopmental toxicity. The *in vivo* results captured thalidomide's well-known teratogenic effects, while not necessarily revealing distinct neurotoxic mechanisms that the DNT battery might be designed to detect. Deltamethrin presented more complex results, particularly in developmental neurotoxicity (DNT) assays. The model-derived HEDs varied significantly depending on the endpoint measured, ranging from 15-fold lower to 80-fold higher than *in vivo* experimental values. These variations align with deltamethrin's known specific effects on neural development (Frank et al., 2017). The compound showed the highest potency in rat neural network formation and oligodendrocyte formation, while requiring higher concentrations to affect human neural network formation and neural crest cell migration. Additional effects on synaptogenesis, neurite development, and radial glia migration were observed at higher concentrations, though these effects were less specific and harder to distinguish from general cytotoxicity (Masjosthusmann et al., 2020). It seems the DNT battery covers

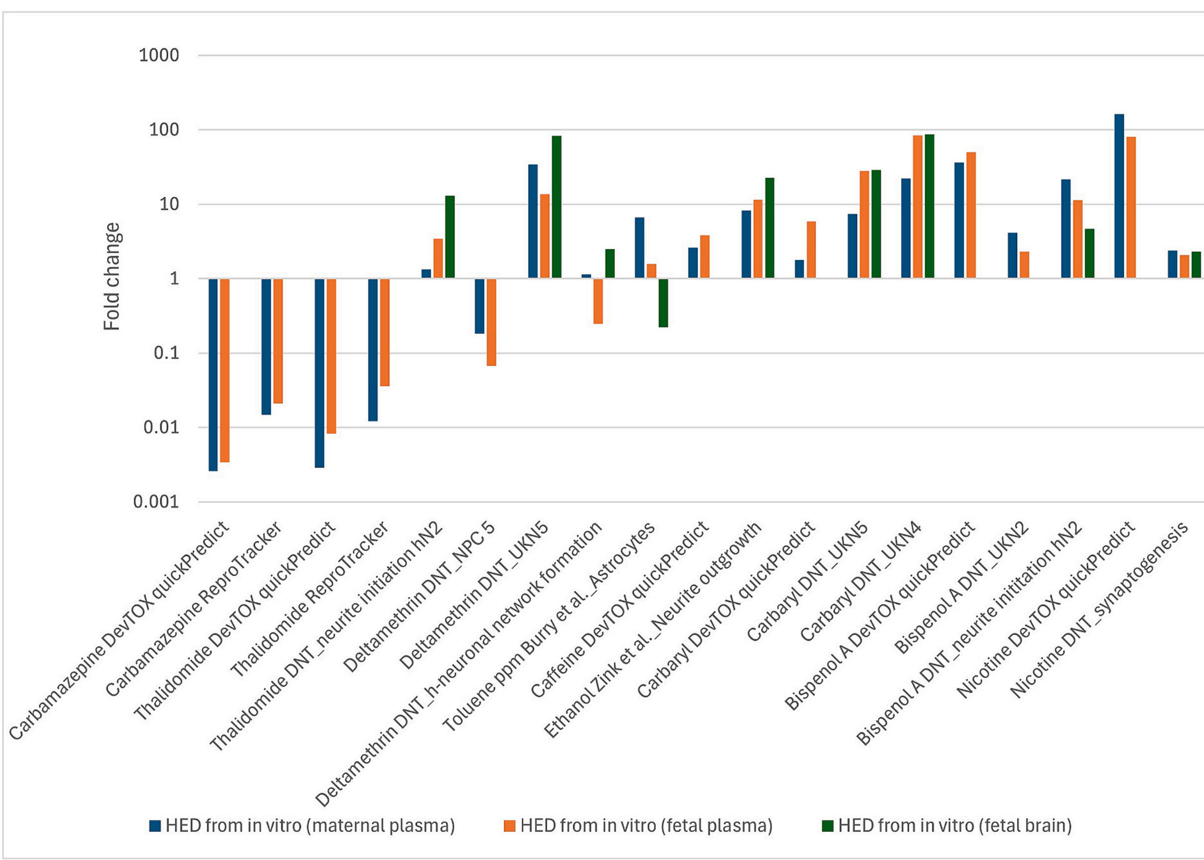

Fig. 6. Fold change between HED from *in vivo* experiments and HED from *in vitro* assays.

deltamethrin mode of action well (voltage gated sodium channels interference). These *in vitro/in vivo* HEDs are several magnitudes higher than the predicted exposure of the general population (1.28E-7 - 7.88E-7 mg/kg/day representing the median and upper bounds for women of reproducible age in the national health and nutrition examination survey (NHANES)).Acetaminophen *in vitro* derived HED from the Stemina DevTOX quickPredict™ assay gives high values (over 10,000 mg/day) which is higher than the therapeutic dose used in the general population (4000 mg/day dosage limit). Acetaminophen is used as a negative control for several DNT studies ((Frank et al., 2017; Harrill et al., 2018; Masjosthusmann et al., 2020) with no effects on the various parameters measured in these studies. However, some experimental studies in mice found some abnormal sperm and reduced birth weight but no embryo or teratogenicity. These differences most likely result from the use of lower acetaminophen concentrations (≤ 100 μM; quantities that are insufficient to generate effects) in these *in vitro* tests (Martin et al., 2022). The doses administered in animal studies often exceed realistic human exposure levels and surpass concentrations used *in vitro* testing, potentially explaining the absence of observable effects in developmental neurotoxicity (DNT) assays. This disconnect between *in vivo* experiments and human exposure scenarios undermines confidence in current safety assessment approaches and highlights the need for more integrated testing strategies that better reflect human biology and actual exposure conditions.

For 6 of the chemicals, the agreement between *in vitro* and *in vivo* HEDs is not good but it is difficult to draw conclusions on the lack of agreement. The comparison between *in vitro* and *in vivo* effects for certain chemicals presents significant challenges in safety assessment. These chemicals often exhibit different types of *in vivo* effects *versus in vitro*, with *in vivo* endpoints sometimes being non-specific, making direct comparisons with specific *in vitro* assays problematic. Ethanol illustrates this complexity perfectly—while it causes general developmental toxicity with ossification effects *in vivo* for example, *in vitro* testing revealed effects on neurite outgrowth. According to ECHA evaluations, developmental neurotoxicity from ethanol occurs only at extremely high doses that don't warrant classification, and there's no specific classification category for neurotoxicity endpoints. ECHA further determined that achieving doses of ethanol sufficient to produce adverse developmental responses would require repeated consumption of large quantities—behavior typically associated with problematic drinking patterns. This reality led ECHA to conclude that classifying ethanol for reproductive or developmental toxicity in the context of chemical substance regulation is neither appropriate nor warranted, highlighting how contextual exposure patterns must inform regulatory decisions beyond simple hazard identification (ECHA). Caffeine *in vitro* derived HED from the Stemina DevTOX quickPredict™ assay (8782–15,569 mg/day) was higher than the safe dose recommended of 400 mg/day and 2 to 4 fold higher than the *in vivo* HED showing teratogenicity in rats. Caffeine as well as carbamazepine did not demonstrate an effect across the DNT assay (Harrill et al., 2018) although being characterized as a DNT positive control. It is unknown why the DNT assays misclassified caffeine and carbamazepine, but it could be a lack of expression of the molecular targets these chemicals affect, a lack of coverage of the biological process targeted by these chemicals, a lack of metabolic competence, or tested concentration ranges below a toxicity threshold (Harrill et al., 2018). Research shows that caffeine only produces gross malformations in rodents when administered as high-dose boluses through intraperitoneal injection or gavage, causing maternal toxicity. One drinking study identified a NOAEL of 205 mg/kg bw/d for teratogenicity, corresponding to a Human Equivalent Dose (HED) of 3398 mg/day. According to OECD's, 2002 report (OECD, March 2002), teratogenic effects require blood caffeine levels exceeding 60 μg/ml, achievable in rodents only through large bolus doses above 40–60 mg/kg bw/day or approximately 800 mg/kg bw/day in drinking water. For context, this would require a 70 kg human to consume about 50 g of caffeine within a short time period.

Nicotine *in vitro* derived HEDs from the Stemina DevTOX quickPredict™ assay gave high values (>10,000 mg/day) whereas for the DNT assays, *in vitro* HEDs vary from 499.5 to 603.2 mg/day (synaptogenesis). Research has established nicotine as an endpoint-selective control, specifically affecting later neurodevelopmental processes like neurite maturation and synaptogenesis while not impacting earlier developmental stages (Aschner et al., 2017; Harrill et al., 2018). This is an example where the DevTOX quickPredict™ is not sensitive to detect specific neurotoxic effects (different window of susceptibility). For chemicals of interest, DNT battery screening allows the reduction and refinement of animal testing strategies.

Bisphenol A *in vitro* derived HED from the Stemina DevTOX quickPredict™ assay was higher than the *in vitro* derived HEDs from the DNT assays. This case illustrates a common limitation in toxicology comparisons, where the *in vivo* experiment's Point of Departure (POD) lacks specificity, encompassing broad developmental and reproductive toxicity endpoints. This makes direct comparison with *in vitro* PODs problematic, as the latter specifically target distinct vulnerability windows such as teratogenicity or neurotoxicity effects. Both *in vivo* and *in vitro* HEDs were several orders of magnitude higher than the *in vivo* predicted HED and the predicted exposure in the population (3.06E-5–3.31E-5 mg/kg/day representing the median and upper bounds for women of reproducible age in NHANES).

Carbaryl *in vitro* derived HED from the Stemina DevTOX quickPredict™ assay was in the same order of magnitude as the DNT assays (1–6 fold difference). Carbaryl demonstrated selective inhibition of neurite growth in the UKN4 assay and showed particular cytotoxicity toward peripheral neurons and mixed NPC cultures. While these cell type-specific effects might have developmental significance (Blum et al., 2023), Health Canada's review found that rat studies showed maternal effects but no clear treatment-related developmental impacts. Observed brain morphometric changes in high-dose groups were deemed not toxicologically significant due to inconsistent patterns between sexes and age groups. Both *in vivo* and *in vitro* HEDs are several orders of magnitude higher than the predicted exposure in the population (4.13E-7–5.12E-5 mg/kg/day representing the median and upper bounds for women of reproducible age in NHANES).

It is important to know that a disagreement between *in vitro* and *in vivo* HEDs does not mean that we can't rely on *in vitro* tests for teratogenicity or developmental neurotoxicity. Understanding the window of susceptibility of the *in vitro* assays they target and how they relate to the *in vivo* experiments is important. Ultimately, confidence levels and uncertainties in PBPK models vary significantly based on available data and assessment stage, necessitating careful evaluation of whether increased model complexity meaningfully improves safety decision-making in pregnancy contexts.

Recent work by Chang et al. (2022) has further validated this approach by comparing various PK/PBPK models for translating *in vitro* developmental concentrations to *in vivo* doses. Their finding of strong correlation between model-generated HEDs and *in vivo* data supports the validity of combining *in vitro* developmental assays with IVIVE for quantitative toxicity estimation. This correlation strengthens the case for using these methods to predict developmental toxicity levels while potentially reducing animal testing requirements.

To advance this research, several key developments are necessary. Better characterization of developmental windows targeted by different assays is essential to ensure appropriate comparison with *in vivo* effects. A framework for meaningfully comparing broad *in vivo* endpoints with specific *in vitro* mechanisms should be developed. The PBPK model parameters require refinement to account for compound-specific characteristics affecting prediction accuracy, particularly regarding maternal-fetal transfer dynamics. Explicit quantification of confidence levels in model predictions needs incorporation into safety assessments. An integrated testing strategy combining complementary *in vitro* assays should be implemented to cover multiple modes of action and developmental windows, guided by adverse outcome pathways. Validation should extend to a broader compound set with well-characterized developmental toxicity profiles to strengthen model confidence. Finally, establishing regulatory acceptance criteria for when PBPK-supported *in vitro* to *in vivo* extrapolation provides sufficient evidence for safety decision-making in pregnancy contexts is crucial.

This work represents important progress toward reducing animal testing while improving the relevance of developmental toxicity assessments to human health protection. The path forward requires refining both the biological understanding of developmental toxicity mechanisms and the computational approaches that translate *in vitro* findings to human risk assessment.

5. Conclusion

The research demonstrates significant progress in physiologically-based pharmacokinetic (PBPK) modeling as a tool to translate *in vitro* developmental toxicity assay results to human equivalent doses (HEDs), refining human risk assessments and potentially reducing the need for animal testing. The combined approach holds promise for improving safety evaluations of chemicals and pharmaceuticals. Our current model requires further validation to determine whether its parameterization and framework can be reliably extended to a broader range of chemicals. Accurate modeling demands a comprehensive understanding of clearance mechanisms to properly simulate the effects of different pregnancy stages. For these models to be applied with confidence, it is essential that the clearance mechanisms are fully elucidated to accurately capture, in simulations, the effects of pregnancy stages. More effort is also needed to address the problem of transporters' involvement.

The development of the three-tiered models (maternal, fetal and refined fetal predictions) was also designed for a tiered modeling approach to PBPK model development. Traditionally, high-quality *in vivo* pharmacokinetic data serves as the gold standard for parameterizing, calibrating, and validating PBPK models. However, such data is often scarce, particularly for human subjects and especially for pregnant women. While some pharmacokinetic data exists for adult populations, data for pregnant women is extremely limited, primarily due to ethical and practical constraints on clinical studies in this sensitive population. As illustrated in Fig. 7, PBPK model development can follow a tiered approach based on available data and the specific requirements of the risk assessment process. At the screening level, a generic PBPK modeling approach utilizes *in silico* and/or *in vitro* ADME data. When *in vivo* pharmacokinetic data becomes available, it enables the development and validation of chemical-specific PBPK models. As regulatory frameworks shift away from traditional animal testing toward new methodologies based on *in vitro* and *in silico* approaches, the field requires innovative strategies to effectively integrate these non-animal data sources. The confidence level and uncertainty in PBPK models will vary depending on the available data and the stage of risk assessment, ranging from early compound development to market-ready risk evaluation.

This evolving landscape necessitates continued refinement of our modeling approaches to maintain accuracy while adapting to new data sources and regulatory requirements.

CRediT authorship contribution statement

Marjory Moreau: Writing – original draft, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Todor Antonijevic:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis. **John Carter Hall:** Visualization, Software, Formal analysis. **Jeffrey Fisher:** Writing – review & editing, Supervision, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial

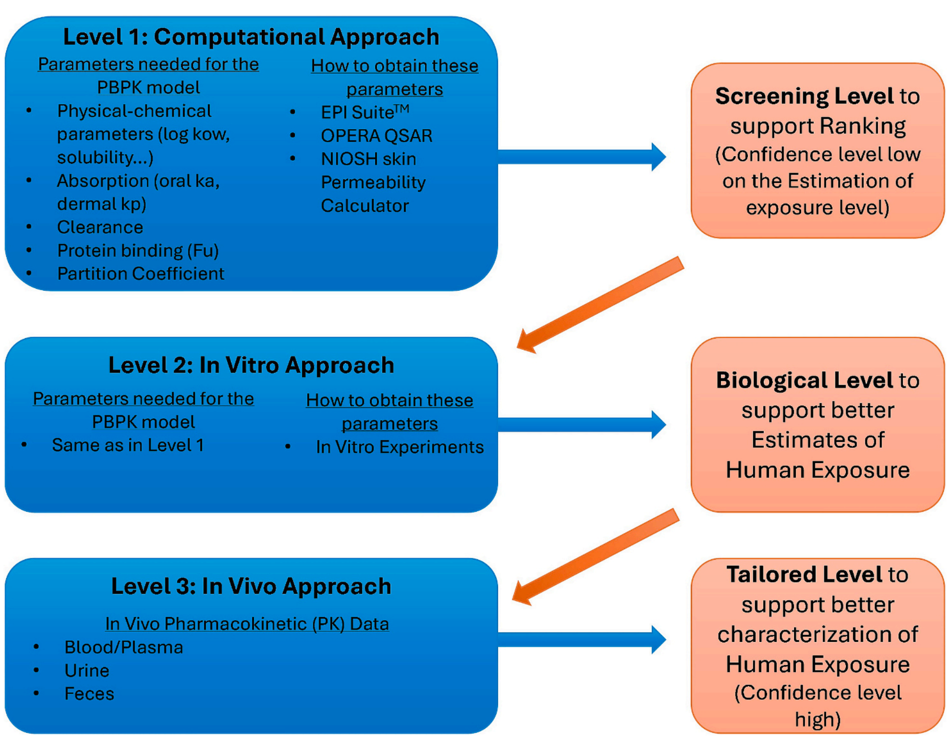

Fig. 7. Tiered testing strategy in PBPK modeling.

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.taap.2025.117435.

Data availability

Most of the data has been published on Mendeley and a link was shared.

Data-Driven Evaluation of PBPK Models for Developmental Toxicity (Original data) (Mendeley Data)

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